

Maximum Likelihood Estimation in the 3-parameter Weibull Distribution A Look through the Generalized Extreme-value Distribution

Hideo Hirose

Faculty of Information Sciences
Hiroshima City University, Asaminami, Hiroshima
Japan

ABSTRACT

The main contribution of this paper is to provide reasonable confidence intervals for maximum likelihood estimates of percentile points in dielectric breakdown voltage probability distributions. The Weibull distributions which include threshold values are often used in breakdown voltage distributions. In some breakdown voltage samples, there exist cases in which the maximum likelihood estimates of all the three Weibull parameters diverge; these cases, in limiting forms, correspond to Gumbel distributions. However, computing confidence intervals of the percentile point estimates using the observed information matrix might be impossible when the distribution models are assumed to be the Weibull type. For such cases, adoption of the generalized extreme-value distribution as an extension of the Weibull distribution allows us to assess this troublesome problem. A Monte Carlo simulation using the newly developed generalized extreme-value distribution parameter estimation code reveals the property of the percentile point estimates for extreme-value type distributions, when the Weibull shape parameter is large. Biases and root mean squared errors of percentile point estimates are investigated both for the maximum likelihood estimates and for some closed form estimates; maximum likelihood estimates provide reasonable confidence intervals. For convenience, a demonstrative example with an estimation algorithm is provided.

1. INTRODUCTION

FOR short-term dielectric breakdown voltage assessment, the Weibull probability distributions often have been used [2–4, 13, 18, 29], and recently the inclusion of the non-negative thresholds to the Weibull distributions have been discussed vigorously [2–4, 18, 29]. The term ‘threshold’ means the location parameter in the 3-parameter Weibull distribution (W3P), and it is not directly relevant to the (long term) life threshold below which breakdown will not occur in this paper. Thus, dealing with the W3P has been becoming more and more important.

It is well known that maximum likelihood parameter estimations in the W3P are tedious and many researchers have long been pursuing algorithms which find the estimates [11, 23, 32]. The probability distribution function of the Weibull distribution is expressed by

$$F^1(x; \eta, \beta, \gamma) = 1 - \exp \left\{ - \left(x - \frac{\gamma}{\eta} \right)^\beta \right\} \quad (1)$$

$$x \geq \gamma, \quad \eta > 0, \quad \beta > 0$$

where η , β and γ are called scale, shape and location parameters, respectively. In estimating the maximum likelihood estimates (MLE) of the three parameters, two difficulties arise in the W3P: one is the ‘non-regular problem’ and the other is the ‘parameter diverging problem’.

Non-regular cases occur when $\beta \leq 2$. There are no MLE when $0 < \beta < 1$; when $1 \leq \beta \leq 2$, MLE exist but are not asymptotically normal [8, 27]. Thus, under such a non-regular condition, maximum likelihood estimation is treated anomalously in the distribution. In order to avoid the difficulty, a variety of methods have been proposed. Among them are the modified moment estimation [6, 7]; modified maximum likelihood estimation [6]; and closed form methods [16, 21, 22, 30–32, 34].

Another annoying property of the maximum likelihood estimation in the W3P is the parameter diverging problem [5, 11]. In some dielectric breakdown voltage tests, the MLE of the Weibull three parameters diverge in the Newton-Raphson process, while the log-likelihood tends to converge. This phenomenon is called the ‘parameter diverging problem’ in this paper. The probability distributions in such cases are seen to be negatively skewed. This paper focuses on this problem. Hirose [11] proposes the use of

Table 1. Dielectric Breakdown Voltage Data of Epoxy Resin.

Data item	Case				
	1	2	3	4	5
1	24.54	27.15	27.66	27.98	28.04
2	28.00	29.13	26.54	27.49	28.57
3	25.69	28.28	26.96	27.85	26.33
4	27.72	27.74	26.15	27.93	29.61
5	28.05	28.87	25.26	24.19	28.17
6	27.53	26.42	29.44	25.01	27.14
7	27.34	24.46	28.32	27.06	29.17
8	26.80	30.88	27.66	27.62	25.44
9	26.51	29.11	28.21	28.94	28.49
10	27.28	27.31	27.80	29.09	27.46
11	28.16	27.54	27.59	27.63	27.31
12	28.86	27.98	26.63	28.28	26.95
13	26.67	28.49	28.08	27.63	27.88
14	28.37	26.25	28.83	28.20	27.30
15	28.37	28.50	27.96	27.95	29.02
16	28.44	25.61	28.13	28.33	29.52
17	28.05	29.50	29.06	27.11	26.89
18	24.61	28.04	26.78	26.47	27.89
19	27.54	27.94	28.00	28.17	28.08
20	26.85	26.66	26.28	27.35	27.75

the 'limiting percentile point estimate' $\hat{x}_p(\eta^{(j)}, \beta^{(j)}\gamma^{(j)})_{(j \rightarrow \infty)}$, in the Newton-Raphson process, and not the parameter estimate itself for such cases; x_p is expressed in (2), and $\eta^{(j)}, \beta^{(j)}$ and $\gamma^{(j)}$ denote the j th values in the Newton-Raphson iterations, while p is the failure probability.

$$x_p = \gamma + \eta |\log(1 - p)|^{1/\beta} \quad (2)$$

This limiting value corresponds to the percentile point estimate in the 2-parameter Gumbel distribution (G2P). Cheng and Iles [5] also recommend the use of the G2P which are called the 'embedded models' associated with the 3-parameter distribution models; in their example, however, the estimated location parameter is negative and not diverging to $-\infty$. They go on to discuss which model between the original 3-parameter model (in their case, $\hat{\gamma} < 0$) and the embedded model should be used by using various hypothesis tests. However, there are highly negatively skewed samples in some experimental cases, for which other distribution models might be more appropriately fitted than the embedded model. The data case 4 in Table 1 and Figure 1 (histograms for Table 1) shows such a case. The coefficient of skewness $\sqrt{\beta_1} = -1.501$ for that case (see Table 1); in the Gumbel distribution, $\sqrt{\beta_1} = -1.14$ [14]. This leads us to introduce an extension of the Weibull distribution to the extreme-value distribution family, i.e. the generalized extreme-value distribution (GEV).

The adoption of the GEV has other benefits. One is that the maximum likelihood parameter estimation in the GEV is easier than in the W3P, which will be discussed in Section 5. References [28] and [5] are relevant to the idea of the parameterization of the GEV model from the W3P model; however, they do not show that using the GEV is effective in estimating the Weibull parameters.

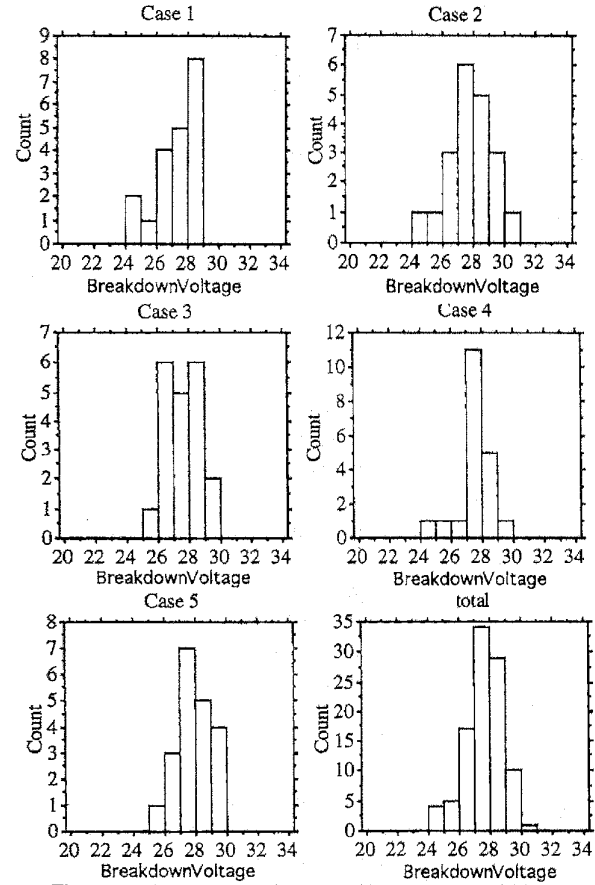


Figure 1. Histograms of sampled data. Same as Table 1.

Hosking [12] developed the estimation code for the GEV MLE in the case of GEV for maxima, but not for the case of GEV for minima, which are dealt with in this paper. The proposed algorithm is an extension of his algorithm. For all cases in the Monte Carlo simulation, this algorithm obtained the MLE of Weibull parameters under the condition that $\beta \geq 1$. The algorithm is shown in Appendix 1.

Secondly, the root mean squared errors (RMSE) can reasonably be simulated when the Weibull shape parameter is large, i.e. the GEV shape parameter is close to zero. The confidence interval of the percentile point estimate cannot be computed by using the observed information matrix (Appendix 2) in the W3P, when Weibull's three parameters diverge. The Monte Carlo simulation supports the use of the percentile point estimate obtained from the GEV model.

As pointed out by Hirose [11], estimates of the Weibull parameters and percentile points by Wyckoff, Bain and Engelhardt's method (WBE) [31] were the best with respect to bias and RMSE, if the the maximum likelihood estimation is excluded; so the comparison of the percentile point estimates is done between the MLE and the closed form method estimates, WBE.

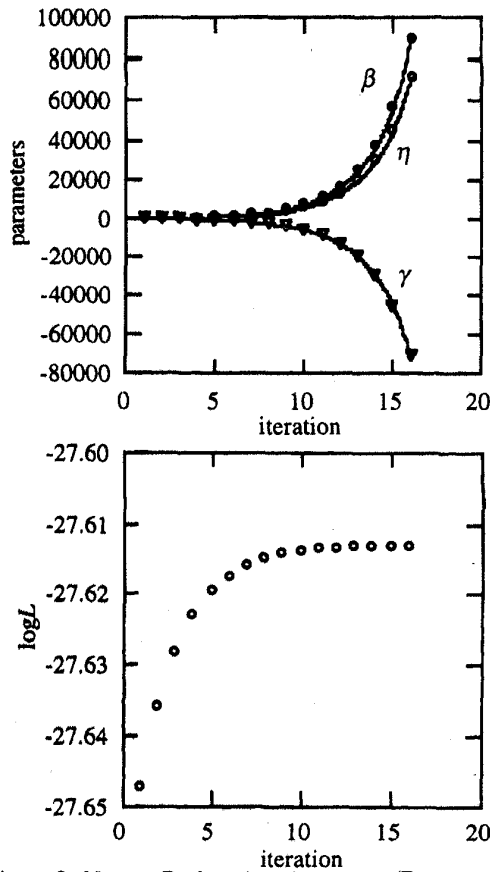


Figure 2. Newton-Raphson iteration process (Data case 4).

2. EXPERIMENTAL DATA IN DIELECTRIC BREAKDOWN VOLTAGE TESTS

Table 1 shows the dielectric breakdown voltages of one hundred epoxy resin test pieces which were tested in the Takaoka Electric Laboratory [11]; but it should be noted that some experiments (e.g. for a generator) are very expensive, so such a large number of test pieces is unusual. Actually, twenty test pieces were tested by applying an ascending voltage to each in turn, and this was repeated five times. Thus, we have five sets of data for the same underlying distribution. Since failure data often fit the Weibull distribution including a certain threshold, it was attempted to obtain the MLE of the 3-parameter Weibull distribution for the five data sets. The computing results using the method described in [11] are shown in Table 2. In the Table, the MLE for 100 data are also provided in the right-hand column. We see that the first and fourth data sets have infinite MLE of parameters but have finite (limiting) log-likelihood and MLE of the percentile point x_p . Figure 2 shows a trace of the Newton-Raphson process for data case 4. In connection with the diverging phenomena in case 1 and 4, it can be seen that highly negatively skewed cases correspond to parameter diverging patterns.

How should we obtain the confidence intervals for the estimates of the percentile point for such cases?

3. REPARAMETERIZATION TO THE GEV MODEL

As mentioned in Sections 1 and 2, there exist cases that the MLE of the parameters in the W3P diverge. In such cases, it seems that $\eta^{(j)}/\beta^{(j)}$ and $\eta^{(j)} + \gamma^{(j)}$ are both converging [11]. These phenomena correspond to data cases 1 and 4 in this paper. The convergence of transformed parameters reminds us of the generalized extreme-value distribution (GEV) which was derived by von Mises [19]. Using the following reparameterization

$$\begin{aligned}\sigma &= \frac{\eta}{\beta} \\ \mu &= \eta + \gamma \\ k &= \frac{1}{\beta}\end{aligned}\quad (3)$$

the probability distribution function of the GEV is expressed by

$$F(x; \sigma, \mu, k) = 1 - \exp \left[- \left\{ 1 + k \left(x - \frac{\mu}{\sigma} \right) \right\}^{1/k} \right] \quad (4)$$

$$\sigma > 0$$

$$1 + k \left(x - \frac{\mu}{\sigma} \right) > 0$$

when $k \neq 0$. The Weibull distribution in (1) corresponds to the case $k > 0$. The cases $k \rightarrow 0$ and $k < 0$ correspond to the Gumbel (5), and Fréchet in (6) distributions, respectively.

$$F^2(x; \sigma, \mu) = 1 - \exp \left\{ - \exp \left(x - \frac{\mu}{\sigma} \right) \right\} \quad (5)$$

$$\sigma > 0$$

$$F^3(x; \eta, \beta, \gamma) = 1 - \exp \left\{ - \left(\gamma - \frac{x}{\eta} \right)^{-\beta} \right\} \quad (6)$$

$$x \leq \gamma \quad \eta > 0 \quad \beta > 0$$

Distributions in (1), (5) and (6) are called extreme-value distributions for the minimum value which were first discovered by Fisher and Tippett [9], and the extreme types theorem was later completely proved by Gnedenko [1, 10]. Historically, the GEV has been mainly treated as a distribution representing the extreme values for maxima [15]; here the extreme-value distributions for minima are discussed.

If we fit the GEV to the fourth data case in Table 1, we can obtain the finite MLE as shown in Table 3. It should be noted that the corresponding value of the maximum log-likelihood ($\log L_{\text{GEV}} = -27.444$) is larger than that obtained by the limiting form of the Weibull (i.e. the Gumbel) distribution $\log L_{\text{W3P}}(\beta \rightarrow \infty) = \log L_{\text{G2P}} = -27.613$. The estimates of the GEV parameters are $\hat{\sigma} = 0.7783$, $\hat{\mu} = 28.032$, $\hat{k} = -0.0811$. Although the standard deviation of the percentile point estimate could not be computed in the Weibull model, the GEV model obtains it because the regularity condition holds when $k < 1/2$ [27]. The computed standard deviation of the 1% point estimate, $\hat{x}_{0.01}$, is 1.697 using the well-known delta method (Appendix 2), if we adopt the

Table 2. Maximum Likelihood estimates of parameters and a percentile point fitted for the Weibull Distribution. $\mu_1 = \frac{1}{n} \sum x_i$, $\mu_2 = \frac{1}{n} \sum (x_i - \mu_1)^2$, $\mu_3 = \frac{1}{n} \sum (x_i - \mu_1)^3$, $\sqrt{\beta_1} = \mu_3 / \mu_2^{3/2}$.

Estimates		Case 1	Case 2	Case 3	Case 4	Case 5	altogether
	$\hat{\eta}$	∞	6.239	5.051	∞	4.725	7.158
	$\hat{\beta}$	∞	4.529	5.267	∞	4.811	6.560
	$\hat{\gamma}$	$-\infty$	22.092	22.921	$-\infty$	23.523	20.916
	$\hat{x}_{0.01}$	23.892	24.352	25.030	24.320	25.339	24.466
Standard Deviation	$\hat{\eta}$		3.396	4.136		3.072	2.354
	$\hat{\beta}$		2.737	4.730		3.479	2.278
	$\hat{\gamma}$		3.283	4.043		2.984	2.321
	$\hat{x}_{0.01}$		0.855	0.787		0.681	0.377
90% Confidence Limits of $\hat{x}_{0.01}$	x_L		22.946	23.735		24.219	23.846
	x_U		25.758	26.325		26.459	25.086
	$\log L$	-28.979	-35.375	-28.652	-27.613	-28.824	-157.073
	$\sqrt{\beta_1}$	-1.032	-0.249	-0.311	-1.501	-0.238	-0.604

GEV. This suggests that it might be better to adopt the GEV as the failure probability distribution function than the W3P alone in a pure likelihood sense for such a case. I do not deeply assess whether we should adopt the G2P or the GEV for this case, although one may think that the difference of the log-likelihood values between those two models is rather small. Adopting the standard deviation computed from the GEV model is considered natural and is supported by the evidence that the approximate upper and lower confidence limits of five data sets all cover the estimate, $\hat{x}_{0.01} = 24.466$, which is computed by using 100 data in Table 1 altogether. The latter property will be investigated by the Monte Carlo simulation study shown in Section 4. However, we cannot simply compute the standard deviation of the percentile point for the limiting Weibull model. For this reason I propose the use of the GEV model.

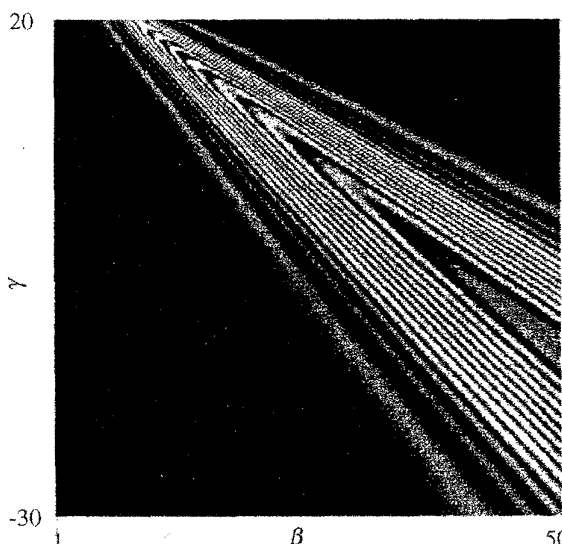


Figure 3. Contour plot of likelihood function for the Weibull distribution based on Data case 4. Scale parameter is optimized.

It is intriguing that the contour plot of the log-likelihood function of the W3P (Figure 3) is very different from that of the GEV

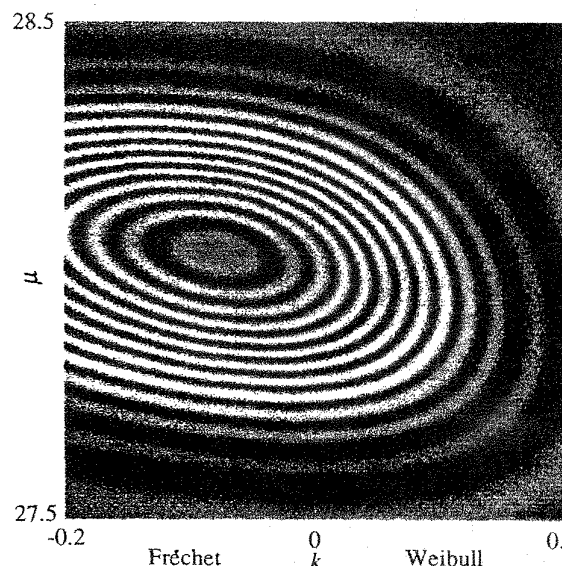


Figure 4. Contour plot of likelihood function for the GEV distribution based on Data case 4. Scale parameter is optimized.

(Figure 4). In the Figures, scale parameters are optimized such that $\partial \log L / \partial \eta = 0$ (in Figure 3) or $\partial \log L / \partial \sigma = 0$ (in Figure 4). The contour plot of the likelihood function of the GEV seems to be smooth around the local maximum point and the location parameter μ does not change the value much even if the shape parameter k varies greatly on a equicontour line. The latter property is very different from the Weibull parameter pattern. The location parameter γ varies a lot as the scale parameter β changes the value. This suggests that by adopting the GEV a search for the local maximum of the likelihood function in the Weibull or Fréchet distribution would be easier. The proposed parameter estimation algorithm uses this property as shown in Appendix 1. A similar property is pointed out by Munro and Wixley [20] for the reparameterized log-normal distribution.

Table 3. Maximum likelihood estimates of parameters and a percentile point fitted for the generalized extreme-value distribution.

Estimates		Case 1	Case 2	Case 3	Case 4	Case 5	altogether
	$\hat{\sigma}$	0.7906	1.3775	0.9589	0.7783	0.9821	1.0913
	$\hat{\mu}$	27.857	28.331	27.972	28.032	28.248	28.074
	$\hat{k}$	-0.1526	0.2208	0.1898	-0.0811	0.2079	0.1525
Standard Deviation	$\hat{x}_{0.01}$	22.584	24.352	25.030	23.692	25.339	24.466
	$\hat{\sigma}$	0.1603	0.2313	0.1724	0.1416	0.1719	0.0797
	$\hat{\mu}$	0.2046	0.3383	0.2416	0.1932	0.2444	0.1188
	$\hat{k}$	0.2058	0.1334	0.1705	0.1493	0.1503	0.0529
90% Confidence Limits of $\hat{x}_{0.01}$	$\hat{x}_{0.01}$	2.735	0.855	0.787	1.697	0.681	0.377
	x_L	18.085	22.946	23.735	20.900	24.219	23.846
	x_U	27.083	25.758	26.325	26.484	26.459	25.086
	$\log L$	-28.672	-35.375	-28.652	-27.444	-28.824	-157.073

Table 4. Number of cases that $\hat{\beta} = 1$ ($\hat{k} = 1$) under the condition $\beta \geq 1$ ($k \leq 1$).

n	β (k)				
	2 (0.5)	3 (0.333)	5 (0.2)	8 (0.125)	100 (0.01)
100	0	0	0	0	0
50	0	0	0	0	0
20	123	43	21	10	2

4. MONTE CARLO SIMULATION

The conditions of the Monte Carlo simulation are as follows. The numbers of sample size $n = 100, 50$ and 20 . The Weibull shape parameter $\beta = 2, 3, 5, 8$ and 100 with the corresponding GEV shape parameter $k = 0.5, 0.333, 0.2, 0.125$ and 0.01 . The Weibull scale and location parameters are β and $-\beta$, e.g. $\sigma = 1$ and $\mu = 0$, respectively. In each case, the number of trials is 1000. The percentile points investigated are $x_{0.01}, x_{0.05}, x_{0.1}, x_{0.5}, x_{0.9}, x_{0.95}$, and $x_{0.99}$.

The cases that the log-likelihood has the maximum value at the corner, $k = 1$ or $\beta = 1$, under the condition, $k \leq 1$ or $\beta \geq 1$, are not considered for assessing the biases and RMSE in this simulation study, but they would not affect the results much because the occurrence ratio of such cases is small except for the case of $n = 20$ and $\beta = 2$, as shown in Table 4.

Comparisons of the estimates of parameters and percentile points are made between the MLE and the WBE, because Hirose [11] mentions that Wyckoff, Bain and Engelhardt's method [31] provides better estimates with respect to bias and RMSE. The WBE of the Weibull parameters are transformed to the estimates in the GEV model for the sake of comparison.

4.1. ESTIMATES OF THE GEV PARAMETERS

The MLE of the Weibull parameters are known to perform badly in comparison to other estimates, e.g. the WBE. In this paper, the MLE in the GEV model are compared with the WBE. Table 5 shows the comparison of the biases between these two methods; the MLE performs well or at least comparably, compared with the WBE except for the cases that $n \leq 50$ and $\beta \leq 2$. Figure 5 shows

Table 5. Biases of estimates of parameters fitted for the generalized extreme-value distribution.

n	β	k	method	σ	μ	k
100	2	0.5	MLE	-0.0018	-0.0097	0.0200
			WBE	-0.0093	-0.0133	0.0132
	3	0.333	MLE	-0.0055	-0.0102	0.0160
			WBE	-0.0033	-0.0364	0.0376
	5	0.2	MLE	-0.0077	-0.0068	0.0096
			WBE	0.0072	-0.0633	0.0664
	8	0.125	MLE	-0.0126	-0.0013	0.0060
			WBE	0.0133	-0.0836	0.0911
	100	0.01	MLE	-0.0100	-0.0036	0.0048
			WBE	0.0411	-0.1349	0.0259
50	2	0.5	MLE	-0.0132	-0.0303	0.0361
			WBE	-0.0047	-0.0225	0.0286
	3	0.333	MLE	-0.0055	-0.0150	0.0271
			WBE	0.0157	-0.0380	0.0582
	5	0.2	MLE	-0.0178	-0.0106	0.0159
			WBE	0.0207	-0.0717	0.0920
	8	0.125	MLE	-0.0141	-0.0172	0.0182
			WBE	0.0362	-0.1016	0.1201
	100	0.01	MLE	-0.0196	-0.0097	0.0105
			WBE	0.0592	-0.1498	0.0546
20	2	0.5	MLE	-0.0448	-0.0337	0.0446
			WBE	-0.0126	-0.0069	0.0357
	3	0.333	MLE	-0.0384	-0.0291	0.0443
			WBE	0.0162	-0.0508	0.1004
	5	0.2	MLE	-0.0439	-0.0296	0.0330
			WBE	0.0388	-0.0941	0.1463
	8	0.125	MLE	-0.0521	-0.0311	0.0285
			WBE	0.0526	-0.1319	0.1846
	100	0.01	MLE	-0.0454	-0.0442	0.0353
			WBE	0.0938	-0.1961	0.1304

the frequency distributions of the estimates of shape parameter $\hat{k}$ as a typical example ($n = 50$ and $\beta = 100$). From the Figure, we can see that almost half of the simulation cases are treated as from the Fréchet model in using the MLE and that all the cases are treated as from the Weibull model in using the WBE. Table 6 shows the comparison of the RMSE of the parameter estimates.

Table 6. RMSE of estimates of parameters fitted for the generalized extreme-value distribution.

n	β	k	method	σ	μ	k
100	2	0.5	MLE	0.0823	0.1151	0.0700
			WBE	0.0826	0.1168	0.0625
	3	0.333	MLE	0.0766	0.1109	0.0694
			WBE	0.0771	0.1200	0.0713
	5	0.2	MLE	0.0743	0.1071	0.0687
			WBE	0.0738	0.1279	0.0895
	8	0.125	MLE	0.0798	0.1145	0.0712
			WBE	0.0808	0.1447	0.1073
	100	0.01	MLE	0.0807	0.1193	0.0775
			WBE	0.0927	0.1838	0.0628
50	2	0.5	MLE	0.1177	0.1564	0.1099
			WBE	0.1171	0.1569	0.0921
	3	0.333	MLE	0.1118	0.1571	0.1083
			WBE	0.1141	0.1599	0.1039
	5	0.2	MLE	0.1092	0.1553	0.1046
			WBE	0.1101	0.1710	0.1209
	8	0.125	MLE	0.1150	0.1601	0.1135
			WBE	0.1209	0.1924	0.1462
	100	0.01	MLE	0.1145	0.1659	0.1134
			WBE	0.1320	0.2260	0.0933
20	2	0.5	MLE	0.1979	0.2501	0.1895
			WBE	0.1913	0.2426	0.1271
	3	0.333	MLE	0.1865	0.2551	0.2043
			WBE	0.1760	0.2484	0.1616
	5	0.2	MLE	0.1907	0.2604	0.2095
			WBE	0.1881	0.2637	0.1938
	8	0.125	MLE	0.1933	0.2639	0.2219
			WBE	0.1943	0.2866	0.2246
	100	0.01	MLE	0.1995	0.2770	0.2288
			WBE	0.2271	0.3329	0.1831

The RMSE of the MLE is nearly the same as that of the WBE.

4.2. ESTIMATES OF THE PERCENTILE POINTS

The performance of the MLE of the percentile points is almost the same as that of the WBE of the percentile points (Table 7). However, it should be noted that the RMSE of the MLE are slightly smaller than those of the WBE when $n \geq 50$ and $\beta \geq 8$ (Table 8), although the RMSE from the WBE is advantageous in a sense that the WBE considers the Fréchet cases as Weibull cases (Figure 5). For convenience, asymptotic standard deviations of percentile point estimates for sample size=1, are provided in Table 9. These values roughly agree with the simulated standard deviation in Table 8 when $n \geq 50$ and $\beta \geq 8$.

The MLE method can compute the approximate standard deviation as well as the approximate confidence interval for the percentile point estimate by using the delta method, when the sample size is large. Table 10 shows the coverage ratios of 90% confidence intervals for percentile point x_p ; short coverage ratios are seen in general from the Table, especially for the low percentile

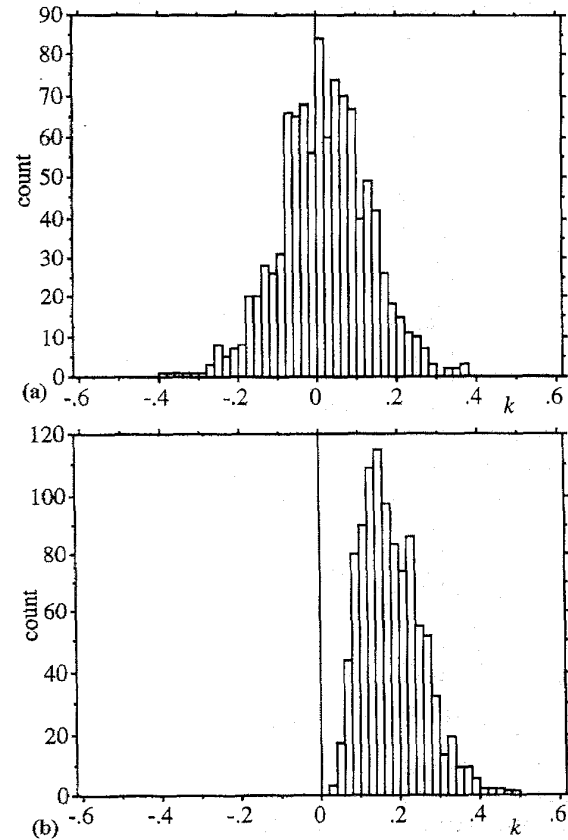


Figure 5. Frequency distribution of estimate k by the Monte Carlo simulation. Random variables are generated from the Weibull model of sample size $n = 50$ and shape parameter $b = 100$ ($k = 0.01$). Number of simulation trials is 1000. (a) Estimated from the MLE. (b) Estimated from the WBE.

points and for the small Weibull shape parameters. However, we can confirm that the asymptotic normality approximately holds for relatively large samples (e.g. 100).

5. DISCUSSIONS

5.1. BASIN OF ATTRACTIONS

A set of starting points from which the Newton-Raphson method finds the solution successfully is called a basin of attractions. A comparison of the volumes of the basin of attractions for the GEV space and for the Weibull space suggests the usefulness of adopting the GEV distribution in iterative processes.

For a simulated data set {0.453929, 0.495161, 0.562135, 0.632766, 0.638778, 0.773269, 0.808026, 0.845425, 0.849381, 0.861406, 0.86719, 0.963768, 0.977251, 0.977928, 1.01967, 1.02241, 1.02453, 1.02731, 1.09366, 1.2665}, two types of cubes are investigated; one is $0.5 \leq \eta \leq 1.5$, $4 \leq \beta \leq 6$, $-0.5 \leq \gamma \leq 0.5$ in the Weibull space, and the other is $0.15 \leq \sigma \leq 0.25$, $0.5 \leq \mu \leq 1.5$, $0.15 \leq k \leq 0.25$ in the GEV space. The maximum point of the log-likelihood function is $(\hat{\eta}, \hat{\beta}, \hat{\gamma}) = (0.9733, 5.0395, -0.0035)$ in the Weibull space and $(\hat{\sigma}, \hat{\mu}, \hat{k}) = (0.1931, 0.1984, 0.9386)$ in the GEV space.

Table 7. Biases of estimates of percentile points fitted for the generalized extreme-value distribution.

n	β	k	method	$x_{0.01}$	$x_{0.05}$	$x_{0.1}$	$x_{0.5}$	$x_{0.9}$	$x_{0.95}$	$x_{0.99}$
100	2	0.5	MLE	0.0407	0.0266	0.0175	-0.0078	-0.0002	0.0097	0.0375
			WBE	0.0337	0.0231	0.0157	-0.0092	-0.0151	-0.0117	0.0008
	3	0.333	MLE	0.0559	0.0351	0.0232	-0.0071	-0.0069	-0.0013	0.0147
			WBE	0.1089	0.0532	0.0258	-0.0328	-0.0221	-0.0084	0.0277
	5	0.2	MLE	0.0494	0.0335	0.0235	-0.0034	-0.0087	-0.0067	0.0001
			WBE	0.2575	0.1046	0.0423	-0.0615	-0.0292	-0.0044	0.0542
	8	0.125	MLE	0.0520	0.0429	0.0336	0.0037	-0.0087	-0.0093	-0.0079
			WBE	0.4444	0.1721	0.0707	-0.0823	-0.0357	-0.0031	0.0717
	100	0.01	MLE	0.0185	0.0295	0.0243	0.0005	-0.0090	-0.0092	-0.0074
			WBE	0.8821	0.2921	0.1008	-0.1402	-0.0463	0.0057	0.1182
50	2	0.5	MLE	0.0734	0.0487	0.0316	-0.0233	-0.0223	-0.0074	0.0389
			WBE	0.0517	0.0322	0.0193	-0.0190	-0.0105	0.0035	0.0437
	3	0.333	MLE	0.0770	0.0509	0.0340	-0.0111	-0.0047	0.0073	0.0406
			WBE	0.1324	0.0601	0.0251	-0.0396	0.0053	0.0355	0.1092
	5	0.2	MLE	0.0816	0.0629	0.0469	-0.0030	-0.0174	-0.0149	-0.0040
			WBE	0.3274	0.1331	0.0538	-0.0730	-0.0137	0.0252	0.1159
	8	0.125	MLE	0.0792	0.0603	0.0419	-0.0107	-0.0194	-0.0149	-0.0003
			WBE	0.5038	0.1833	0.0637	-0.1065	-0.0198	0.0309	0.1451
	100	0.01	MLE	0.0310	0.0574	0.0469	-0.0017	-0.0199	-0.0198	-0.0152
			WBE	0.9859	0.3249	0.1087	-0.1597	-0.0332	0.0340	0.1788
20	2	0.5	MLE	0.1124	0.1006	0.0804	-0.0151	-0.0456	-0.0310	0.0286
			WBE	0.0953	0.0733	0.0565	0.0001	0.0044	0.0222	0.0768
	3	0.333	MLE	0.1207	0.1159	0.0916	-0.0121	-0.0340	-0.0179	0.0386
			WBE	0.2452	0.1292	0.0703	-0.0497	0.0153	0.0657	0.1932
	5	0.2	MLE	0.0685	0.1140	0.0949	-0.0110	-0.0444	-0.0357	0.0002
			WBE	0.4724	0.2033	0.0892	-0.0979	0.0081	0.0768	0.2395
	8	0.125	MLE	0.0114	0.1152	0.1024	-0.0097	-0.0544	-0.0496	-0.0225
			WBE	0.7151	0.2821	0.1140	-0.1381	-0.0048	0.0769	0.2653
	100	0.01	MLE	-0.0564	0.1127	0.0979	-0.0246	-0.0592	-0.0513	-0.0206
			WBE	1.2373	0.4184	0.1406	-0.2129	-0.0135	0.0933	0.3284

To the first cube, 98 points out of equally divided 11^3 grid points in the cube are successful in the Newton-Raphson iteration by the Weibull approach, while 150 points are successful by the GEV approach; all the points in the Weibull space are converted to the points in the GEV space. On the other hand, 213 points out of equally divided 11^3 grid points are successful by the GEV approach to the second cube, while 148 points are successful by the Weibull approach. These two superior success ratios by the GEV approach suggest that a search for the solution of the maximum likelihood equations in the GEV space is better than in the Weibull space. Figure 6 shows two basins of attractions by the two approaches to the first cube. Thus, the use of the GEV distribution is not a mere transformation.

5.2. DERIVATION OF THE EXTREME-VALUE TYPE DISTRIBUTIONS

It is well known that the Weibull model is connected to the weak link-chain breakage problem. In addition, the Weibull distribution is strongly related to the other two types of extreme-value distributions, the Gumbel and the Fréchet distributions.

A brief description of the derivation of the extreme-value distribution is explained as follows [17]. If $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ are n samples from a cumulative distribution function (cdf) $F(x)$, then $G_n(x) = F^n(x)$ becomes a *idf* associated with $x_{(n)}$ which is the maximum value in n samples. Assume that $n = km$ and $m, n \rightarrow \infty$ with k fixed. If $x_{(n)}$ has the limiting form when $n \rightarrow \infty$ under such a condition, the function $G(x) = G_n(x)$ should follow (7).

$$G^k(x) = G(a_k x + b_k) \quad (7)$$

From this functional equation, three types of asymptotic *cdf* are derived as follows.

Gumbel type

$$G(x) = \exp[-\exp\{-nF'(x_{on})(x - x_{on})\}] \quad (8)$$

Fréchet type

$$G(x) = \exp[-(x/x_{on})^{-k}] \quad (9)$$

Weibull type

$$G(x) = \exp[-\{(-nF^{(k)}(x_u)/k!)^{1/k}(x - x_u)\}^k] \quad (10)$$

Table 8. RMSE of estimates of percentile points fitted for the generalized extreme-value distribution.

n	β	k	method	$x_{0.01}$	$x_{0.05}$	$x_{0.1}$	$x_{0.5}$	$x_{0.9}$	$x_{0.95}$	$x_{0.99}$
100	2	0.5	MLE	0.0930	0.0769	0.0787	0.1077	0.1658	0.2069	0.3190
			WBE	0.0899	0.0771	0.0787	0.1074	0.1703	0.2084	0.3085
	3	0.333	MLE	0.1819	0.1257	0.1137	0.1117	0.1361	0.1653	0.2467
			WBE	0.1937	0.1324	0.1163	0.1181	0.1436	0.1692	0.2448
	5	0.2	MLE	0.3181	0.1890	0.1521	0.1116	0.1203	0.1411	0.1999
			WBE	0.3712	0.2064	0.1551	0.1293	0.1294	0.1451	0.2064
	8	0.125	MLE	0.4472	0.2450	0.1895	0.1227	0.1193	0.1371	0.1891
			WBE	0.5503	0.2782	0.1966	0.1501	0.1293	0.1409	0.2030
	100	0.01	MLE	0.7318	0.3434	0.2419	0.1306	0.1111	0.1220	0.1605
			WBE	0.9841	0.4037	0.2481	0.1966	0.1225	0.1237	0.2011
50	2	0.5	MLE	0.1545	0.1207	0.1161	0.1470	0.2313	0.2957	0.4740
			WBE	0.1410	0.1171	0.1144	0.1443	0.2376	0.2966	0.4519
	3	0.333	MLE	0.2763	0.1843	0.1624	0.1582	0.1929	0.2374	0.3650
			WBE	0.2676	0.1869	0.1637	0.1582	0.2022	0.2471	0.3746
	5	0.2	MLE	0.4707	0.2716	0.2150	0.1594	0.1801	0.2119	0.3009
			WBE	0.4800	0.2800	0.2157	0.1726	0.1890	0.2206	0.3219
	8	0.125	MLE	0.6656	0.3522	0.2668	0.1708	0.1718	0.2017	0.2882
			WBE	0.6781	0.3607	0.2674	0.2031	0.1803	0.2094	0.3216
	100	0.01	MLE	1.1038	0.4924	0.3419	0.1817	0.1567	0.1739	0.2318
			WBE	1.1496	0.5094	0.3387	0.2462	0.1633	0.1805	0.2962
20	2	0.5	MLE	0.3368	0.2212	0.2068	0.2364	0.3777	0.4840	0.7729
			WBE	0.2440	0.2110	0.2017	0.2208	0.3886	0.4854	0.7264
	3	0.333	MLE	0.5807	0.3132	0.2633	0.2550	0.3221	0.4025	0.6371
			WBE	0.4273	0.3041	0.2612	0.2409	0.3317	0.4077	0.6198
	5	0.2	MLE	1.1061	0.4729	0.3591	0.2729	0.2892	0.3516	0.5414
			WBE	0.7085	0.4452	0.3573	0.2725	0.3015	0.3703	0.5790
	8	0.125	MLE	1.4449	0.6006	0.4296	0.2822	0.2697	0.3217	0.4913
			WBE	0.9691	0.5537	0.4230	0.3052	0.2756	0.3321	0.5349
	100	0.01	MLE	2.2874	0.8664	0.5774	0.3042	0.2573	0.2952	0.4329
			WBE	1.5425	0.7716	0.5568	0.3711	0.2589	0.3151	0.5464

Table 9. Asymptotic standard deviations of percentile point estimates.

β	k	$x_{0.01}$	$x_{0.05}$	$x_{0.1}$	$x_{0.5}$	$x_{0.9}$	$x_{0.95}$	$x_{0.99}$
3	0.333	1.4289	1.1519	1.1077	1.0707	1.3896	1.6547	2.3240
5	0.2	2.8780	1.8216	1.5278	1.1415	1.2433	1.4405	1.9695
8	0.125	4.1059	2.3361	1.8328	1.1800	1.1674	1.3303	1.7899
100	0.01	6.8862	3.3789	2.4202	1.2374	1.0599	1.1754	1.5409

In the third type, $F(x)$ is assumed to have a terminal point x_u and that $F^{(r)}(x_u) = 0$, ($r = 1, 2, \dots, k-1$), and $F^{(k)} \neq 0$. By regarding $-x_{(1)}$ as $x_{(n)}$, the cdf for $x_{(1)}$ is obtained by a transformation of $H(x) = 1 - G(-x)$, and we can obtain three types of extreme-value distributions for minima as shown in Equations (5), (6), and (1), respectively.

Since we now know that the three types of extreme-value distributions have the same origin, it is natural to consider the expanded model, the GEV, from the Weibull model.

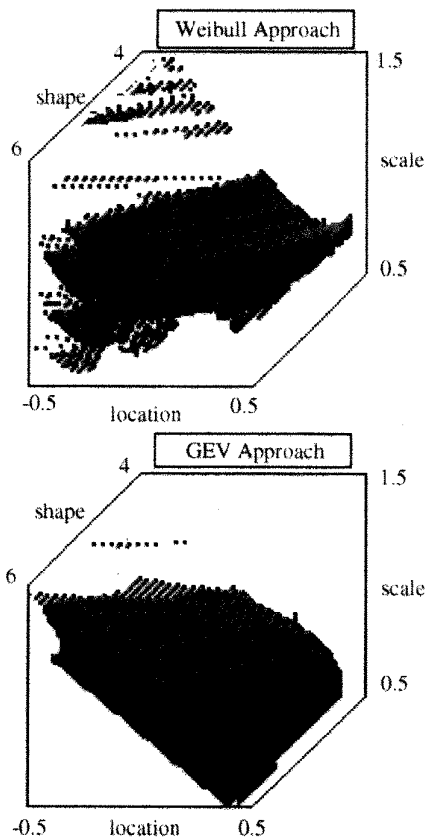
5.3. MODEL SELECTION

One might think that adopting the other probability distribution models are more appropriate to assess the confidence interval

of a percentile point in a breakdown voltage phenomenon, e.g. the 2-parameter Weibull distribution and the Gumbel distribution. This model selection problem is important but too difficult to go into in the discussion. Such a discussion is left as an open question here. Only approximate confidence intervals of the percentile point $x_{0.01}$ for the two parameter Weibull distribution ($\gamma = 0$ is fixed in (1)) and the Gumbel distribution are provided here. Approximate 90% confidence intervals of $x_{0.01}$ are [20.900, 26.484] for the GEV, [23.424, 25.561] for the W2P and [23.125, 25.515] for the G2P; in all the three cases, $\hat{x}_{0.01} = 24.466$ from a 100 sample GEV estimate is included in those confidence intervals. The MLE of the parameters in the G2P and in the W2P is ($\hat{\sigma} = 0.7994$, $\hat{\mu} = 27.997$), and ($\hat{\eta} = 27.984$, $\hat{\beta} = 34.519$), respectively; the value of the log-likelihood in the W2P is -27.770 .

Table 10. Coverage ratios of 90% confidence intervals for the MLE of percentile points x_p .

n	β	k	$x_{0.01}$	$x_{0.05}$	$x_{0.1}$	$x_{0.5}$	$x_{0.9}$	$x_{0.95}$	$x_{0.99}$
100	2	0.5	0.792	0.841	0.858	0.872	0.904	0.912	0.909
	3	0.333	0.824	0.847	0.864	0.886	0.906	0.906	0.910
	5	0.2	0.862	0.875	0.888	0.906	0.905	0.904	0.896
	8	0.125	0.863	0.860	0.875	0.889	0.882	0.878	0.874
	100	0.01	0.871	0.871	0.881	0.869	0.886	0.890	0.898
50	2	0.5	0.733	0.780	0.834	0.882	0.889	0.891	0.903
	3	0.333	0.801	0.829	0.852	0.876	0.897	0.904	0.906
	5	0.2	0.835	0.851	0.867	0.901	0.873	0.885	0.896
	8	0.125	0.820	0.839	0.853	0.884	0.876	0.878	0.881
	100	0.01	0.826	0.849	0.861	0.875	0.879	0.885	0.883
20	2	0.5	0.692	0.715	0.773	0.875	0.843	0.855	0.865
	3	0.333	0.726	0.741	0.778	0.880	0.868	0.878	0.878
	5	0.2	0.757	0.767	0.800	0.857	0.859	0.863	0.861
	8	0.125	0.754	0.774	0.811	0.869	0.839	0.844	0.852
	100	0.01	0.734	0.760	0.787	0.859	0.831	0.834	0.843

Figure 6. Basins of attractions by the two approaches: the Weibull and the GEV. $0.5 < \text{scale} < 1.5$, $4 < \text{shape} < 6$, $-0.5 < \text{location} < 0.5$.

6. CONCLUDING REMARKS

For some experimental samples in electrical engineering, there exist cases in which the MLE of Weibull's three parameters diverge. This paper considered the treatment of confidence intervals of the percentile point estimates for such cases. Adoption

of the generalized extreme-value distribution when the Weibull shape parameter is large proved to provide reasonable RMSE for the MLE of percentile points from the Monte Carlo simulation study. In addition, the paper showed the newly developed effective maximum likelihood parameter estimation scheme in the generalized extreme-value distribution model, and the maximum likelihood estimation of parameters are successfully obtained under conditions in which the Weibull shape parameter is ≥ 1.0 .

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7. APPENDIX 1

7.1. MAXIMUM LIKELIHOOD PARAMETER ESTIMATION IN THE GEV MODEL

The main idea of the proposed algorithm lies on a assumption that $\hat{\sigma}(k)$ and $\hat{\mu}(k)$ do not vary as much as k does; $\hat{\sigma}(k)$ and $\hat{\mu}(k)$ denote the MLE for the GEV distribution when k is fixed. Moreover, when $k \rightarrow 0$, which corresponds to the Gumbel distribution, we can easily estimate the MLE of parameters in the Gumbel distribution by using the Newton-Raphson method. Since the MLE of parameters when using the Newton-Raphson method are very sensitive in general with respect to the initial guesses, the following method is one of the recommended methods for a stable and successful searching scheme of the MLE.

Step 1: Obtain the sample's mean and standard deviation as follows

$$m = \frac{1}{n} \sum_{i=1}^n x_i \quad (11)$$

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - m)^2 \quad (12)$$

where n denotes the number of samples. Set initial guesses, $\sigma_0^{(0)}$ and $\mu_0^{(0)}$, for the MLE of $\hat{\mu}(\text{G2P})$ and $\hat{\sigma}(\text{G2P})$ as

$$\sigma_0^{(0)} = \sqrt{\frac{6}{\pi}}s = 0.779697s \quad (13)$$

$$\mu_0^{(0)} = m + 0.577216\sigma \quad (14)$$

Obtain the MLE of G2P using $\sigma_0^{(0)}$ and $\mu_0^{(0)}$. Set $\sigma_0 = \hat{\sigma}(\text{G2P})$, $\mu_0 = \hat{\mu}(\text{G2P})$, and $k_0 = 0$.

Step 2: Define a small number $\Delta k > 0$, say $\Delta k = 0.01$. Set $k_1 = \Delta k$, and $k_{-1} = -\Delta k$. Obtain the MLE of the 2-parameter GEV distribution under the constraint of $k = k_1$, by using σ_0 and μ_0 as initial guesses. Set these MLE as σ_1 and μ_1 . Obtain σ_{-1} and μ_{-1} corresponding to k_{-1} in a very similar manner.

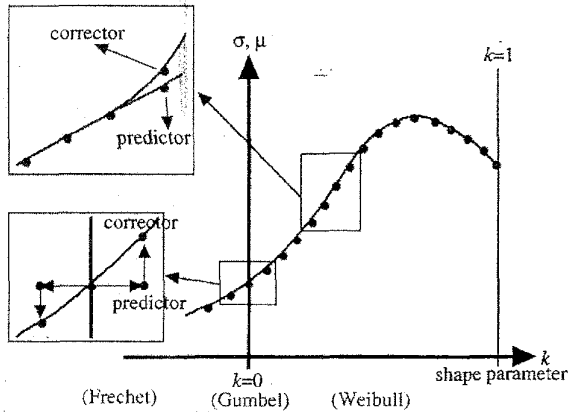


Figure 7. Maximum likelihood parameter estimation scheme in the GEV space.

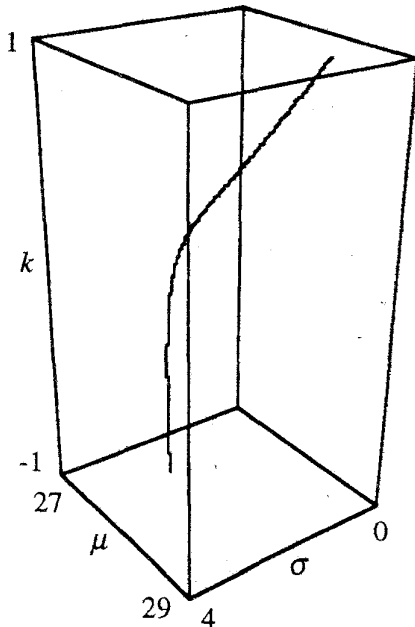


Figure 8. The curve which consist of local maximum points $(\sigma(k), \mu(k), k)$.

Step 3: By using the three points, (σ_{-1}, k_{-1}) , (σ_0, k_0) , and (σ_1, k_1) , obtain an extrapolated point $(\sigma_2^{(0)}, k_2)$ by solving a quadratic equation, where $k_2 = k_1 + \Delta k$ (Figure 7). Obtain $(\mu_2^{(0)}, k_2)$ in a similar manner. By using $(\sigma_2^{(0)}, \mu_2^{(0)})$ as initial guesses, obtain the MLE, σ_2 and μ_2 of the 2-parameter GEV distribution under the constraint of $k = k_2$. Assuming that $(\sigma_{\nu-2}, \mu_{\nu-2})$, $(\sigma_{\nu-1}, \mu_{\nu-1})$ and $(\sigma_{\nu}, \mu_{\nu})$ have already been obtained by the previous steps, then $\sigma_{\nu+1}$ and $\mu_{\nu+1}$ for $k_{\nu+1} = k_{\nu} + \Delta k$ are obtained by the Newton-Raphson method using $\sigma_{\nu+1}^{(0)}$ and $\mu_{\nu+1}^{(0)}$ as the initial values (Figure 7). $\sigma_{\nu+1}^{(0)}$ is an extrapolated value from $(\sigma_{\nu-2}, k_{\nu-2})$, $(\sigma_{\nu-1}, k_{\nu-1})$ and (σ_{ν}, k_{ν}) ; $\mu_{\nu+1}^{(0)}$ is similarly obtained. If the constraint, $\sigma_{\nu+1}^{(j)} > 0$ or $1 + k_{\nu+1}^{(j)}\{(x_i - \mu_{\nu+1}^{(j)})/\sigma_{\nu+1}^{(j)}\} > 0$, is violated in the Newton-Raphson iteration, we use an alternative extrapolated point which is closer to the point $(\sigma_{\nu}, \mu_{\nu}, k_{\nu})$; j denotes the Newton-Raphson iteration count for the point $(k_{\nu+1}, \sigma_{\nu+1}, \mu_{\nu+1})$. If Δk is small, then convergence would be confirmed. This continues until $k = 1 - \Delta k$. Special treatment is needed at the corner, $k = 1$, as Rockette et al. have pointed out [26].

Step 4: A similar process to Step 3 is taken in the opposite direction from $k = k_{-2}$ to $k = k^*$, where usually $k^* = -1$.

Step 5: Search for a maximum point of $\log L(\sigma(k_{\nu}), \mu(k_{\nu}), k_{\nu})$ from all the knots which we are investigating. If $k_{\nu} \neq 1$ and $k_{\nu} \neq k^*$, then do the 3-parameter Newton-Raphson iteration using the initial values σ_{ν}, μ_{ν} and k_{ν} (if $\nu = 0$, use the next point (σ_1, μ_1, k_1)). The converged variables may be the (local) MLE. We must check if this point is not a saddle point by using the determinants of the submatrices formed by the observed information matrix (Appendix 2); the log-likelihood at this point should also be checked if it is larger than the log-likelihood at the corner points, $k = 1$ and $k = k^*$. If $\log L(\sigma(k = 1), \mu(k = 1), 1)$ is still maximum, the MLE of the Weibull parameters are $\hat{\gamma} = x_{\min}$, $\hat{\beta} = 1$ and $\hat{\eta} = \frac{1}{n} \sum_{i=1}^n (x_i - x_{\min})$ under the constraint that $k \leq 1$; $x_{\min}$ is the minimum observation in the experiment. If $\log L(\sigma(k^*), \mu(k^*), k^*)$ is maximum, k^* should be set much smaller.

In these steps, the first and second derivatives of $\log L$ and Fisher's information matrix are obtained in a similar manner in [24, 25], but the specific formulation is given in Appendix 2 for convenience in programming. As mentioned in the first paragraph of this Appendix, the curve which is constructed by connecting $(\sigma(k_{\nu}), \mu(k_{\nu}), k_{\nu})$ points is shown in Figure 8, for the data in case 4. We can confirm that the location parameter μ does not change value much with variations in the shape parameter k in the estimation process, which suggests a more stable search for the solution in the GEV space than in the Weibull space.

8. APPENDIX 2

8.1. NEWTON RAPHSON ALGORITHM

8.1.1. GENERAL

Denote the unknown parameters by $\theta = (\theta_1, \theta_2, \theta_3, \dots, \theta_l)^T$, and the log-likelihood function by $\log L(\theta_1, \theta_2, \theta_3, \dots, \theta_l)$. The Hessian matrix of the log-likelihood function is

$$H(\theta) = \left(\partial^2 \log \frac{L}{\partial \theta_a \partial \theta_b} \right) \quad (15)$$

The matrix $-(\partial^2 \log L / \partial \theta_a \partial \theta_b)$ is called the 'observed information matrix'. Then, the solution of the likelihood equations

$$\partial \log \frac{L}{\partial \theta_a} = 0 \quad a = 1, 2, \dots, l \quad (16)$$

can be obtained successively by

$$\theta^{(j+1)} = \theta^{(j)} - H(\theta^{(j)})^{-1} F(\theta^{(j)}) \quad (17)$$

until $\max |\theta_a^{(j+1)} - \theta_a^{(j)}| < \varepsilon$, where

$$F(\theta) = \left(\partial \log \frac{L}{\partial \theta_1}, \partial \log \frac{L}{\partial \theta_2}, \dots, \partial \log L \partial \theta_l \right)^T \quad (18)$$

When using 2-parameter estimation such as the Gumbel distribution and the 2-parameter GEV distribution under the constraint of fixed k , use θ_1 and θ_2 ; in the case of 3-parameter estimation, use θ_3 in addition.

8.1.2. HESSIAN MATRIX FORM FOR THE GUMBEL DISTRIBUTION

The log-likelihood function for the Gumbel distribution is

$$\log L = \sum_{i=1}^n \{-\log \mu + z_i - \exp(z_i)\} \quad (19)$$

where $z_i = (x_i - \mu)/\sigma$. The function $F(\theta)$ is expressed as

$$\partial \log \frac{L}{\partial \mu} = -\frac{1}{\sigma} \sum_{i=1}^n \{1 - \exp(z_i)\} \quad (20)$$

$$\partial \log \frac{L}{\partial \sigma} = -\frac{1}{\sigma} \sum_{i=1}^n [1 + z_i \{1 - \exp(z_i)\}] \quad (21)$$

Thus, the components of the Hessian matrix are

$$\partial^2 \log L \partial \mu^2 = -\frac{1}{\sigma^2} \sum_{i=1}^n \exp(z_i) \quad (22)$$

$$\partial^2 \log \frac{L}{\partial \mu \partial \sigma} = \frac{1}{\sigma^2} \sum_{i=1}^n \{1 - (1 + z_i) \exp(z_i)\} \quad (23)$$

$$\partial^2 \log \frac{L}{\partial \sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n \{1 + [2z_i \{1 - \exp(z_i)\} - z_i^2 \exp(z_i)]\} \quad (24)$$

8.1.3. HESSIAN MATRIX FORM FOR THE GEV DISTRIBUTION

The log-likelihood function for the GEV distribution is

$$\log L = \sum_{i=1}^n (-\log \sigma - \log z_i + \log \phi_i - \phi_i) \quad (25)$$

where $z_i = 1 + k\{(x_i - \mu)/\sigma\} = 1 + ky_i$, and $\phi_i = z_i^{1/k}$. The function $F(\theta)$ is expressed as

$$\partial \log \frac{L}{\partial \mu} = \sum_{i=1}^n \frac{1}{\sigma z_i} \psi_i \quad (26)$$

$$\partial \log \frac{L}{\partial \sigma} = -\sum_{i=1}^n \frac{1}{\sigma} \left(1 - \frac{y_i \psi_i}{z_i} \right) \quad (27)$$

$$\partial \log \frac{L}{\partial k} = -\sum_{i=1}^n \frac{1}{k} \left\{ \frac{y_i \psi_i}{z_i} + (1 - \phi_i) \log \phi_i \right\} \quad (28)$$

where $\psi_i = k - 1 + \phi_i$.

The second derivatives of the likelihood function are complicated to write out. So, only the relevant equations for the chain rule are shown as follows

$$\partial y_i / \partial \mu = -1/\sigma \quad (29)$$

$$\partial y_i / \partial \sigma = -y_i/\sigma \quad (30)$$

$$\partial y_i / \partial k = 0 \quad (31)$$

$$\partial z_i / \partial \mu = -k/\sigma \quad (32)$$

$$\partial z_i / \partial \sigma = -ky_i/\sigma \quad (33)$$

$$\partial z_i / \partial k = y_i \quad (34)$$

$$\partial \phi_i / \partial \mu = -\phi_i/(\sigma z_i) \quad (35)$$

$$\partial \phi_i / \partial \sigma = -y_i \phi_i/(\sigma z_i) \quad (36)$$

$$\partial \phi_i / \partial k = \phi_i(-\log \phi_i + y_i/z_i)/k \quad (37)$$

$$\partial \psi_i / \partial \mu = \partial \phi_i / \partial \mu \quad (38)$$

$$\partial \psi_i / \partial \sigma = \partial \phi_i / \partial \sigma \quad (39)$$

$$\partial \psi_i / \partial k = 1 + \partial \phi_i / \partial k \quad (40)$$

When using the 2-parameter estimation under the constraint that k is fixed, discard the terms and equations relevant to k .

8.1.4. DELTA METHOD

For obtaining the approximate confidence intervals of certain functions of $\hat{\theta}$, the delta method is first used because of its simple form. The approximate standard error $e(g(\hat{\theta}))$ of $g(\hat{\theta})$ is computed as

$$e(g(\hat{\theta})) \approx \sum_a \sum_b \frac{\partial g(\hat{\theta})}{\partial \theta_a} \frac{\partial g(\hat{\theta})}{\partial \theta_b} \text{Cov}(\hat{\theta}_a, \hat{\theta}_b) \quad (41)$$

where $\text{Cov}(\hat{\theta}_a, \hat{\theta}_b)$ is the covariance computed from the observed information matrix.

9. APPENDIX 3

9.1. NUMERICAL EXAMPLE IN THE ESTIMATION

A numerical example is demonstrated here for data case 2 in Table 1.

For the Gumbel distribution, $m = 27.793$ and $s = 1.42484$; so, $\sigma_0^{(0)} = 1.11094$ and $\mu_0^{(0)} = 28.4343$ are the first guesses. Then, $\sigma_0 = \hat{\sigma}(\text{G2P}) = 1.36428$, $\mu_0 = \hat{\mu}(\text{G2P}) = 28.4911$ are obtained by the Newton-Raphson method.

We set $\Delta k = 0.01$. Then, $\sigma_1 = 1.35949$, $\mu_1 = 28.4854$, $\sigma_{-1} = 1.36943$ and $\mu_{-1} = 28.4966$ are estimated corresponding to $k = 0.01$ and $k = -0.01$ respectively.

By using three points, (σ_{-1}, k_{-1}) , (σ_0, k_0) and (σ_1, k_1) , an extrapolated point $(\sigma_2^{(0)}, k_2) = (1.35507, 0.02)$ is obtained; similarly a point $(\mu_2^{(0)}, k_2) = (28.4797, 0.02)$ is obtained. Then, $(\sigma_2, \mu_2) = (1.35507, 28.4797)$ is estimated. In a similar manner, an opposite point $(\sigma_{-2}, \mu_{-2}) = (1.37493, 28.5020)$ is also estimated by the initial guesses $(\mu_{-2}^{(0)}, \mu_{-2}^{(0)}) = (1.37496, 28.5020)$.

After obtaining the log-likelihood at the points $(\sigma_\nu, \mu_\nu, k_\nu)$ $(-1 \leq k_\nu \leq 0.99)$, an optimum grid point $(\sigma_{22}, \mu_{22}, k_{22}) = (1.37688, 28.3321, 0.22)$ is obtained with $\log L = -35.3748$.

Using these values, the local maximum point $(\hat{\sigma}, \hat{\mu}, \hat{k}) = (1.37754, 28.3314, 0.22079)$ is estimated. The maximum log-likelihood value is -35.3747 . By converting GEV parameters to the Weibull ones, $(\hat{\eta}, \hat{\beta}, \hat{\gamma}) = (6.23894, 4.52906, 22.0924)$ is obtained.

The observed information matrix for the GEV distribution is

$$V = - \begin{bmatrix} \frac{\partial^2 \log L}{\partial \sigma^2} & \frac{\partial^2 \log L}{\partial \mu \partial \sigma} & \frac{\partial^2 \log L}{\partial k \partial \sigma} \\ \frac{\partial^2 \log L}{\partial \sigma \partial \mu} & \frac{\partial^2 \log L}{\partial \mu^2} & \frac{\partial^2 \log L}{\partial k \partial \mu} \\ \frac{\partial^2 \log L}{\partial \sigma \partial k} & \frac{\partial^2 \log L}{\partial \mu \partial k} & \frac{\partial^2 \log L}{\partial k^2} \end{bmatrix} \quad (42)$$

$$= \begin{bmatrix} 24.5712 & -1.88119 & -21.9042 \\ -1.88119 & 10.1881 & 10.8187 \\ -21.9042 & 10.8187 & 83.9127 \end{bmatrix}$$

and V^{-1} is

$$V^{-1} = \begin{bmatrix} 0.0534973 & -0.00568966 & 0.0147564 \\ -0.00568966 & 0.114415 & -0.0162233 \\ 0.0147564 & -0.0162233 & 0.0177976 \end{bmatrix} \quad (43)$$

Using the asymptotic normality, approximated 90% confidence intervals for $\hat{\sigma}, \hat{\mu}, \hat{k}$ are calculated as $[0.997, 1.758]$, $[27.78, 28.89]$, $[0, 0.44]$.

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